

Mourad Heddaya

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Experience building evaluation and alignment systems for LLMs. Built and deployed AI systems with automated pipelines for detecting hallucination, bias, and unsafe model behavior. Background in reasoning compression, long-context and multi-turn evaluation, and large-scale data processing.

Education

Ph.D. in Computer Science, University of Chicago, 2021–2026 (expected May).

Advisor: Chenhao Tan

B.S. in Informatics, University of Washington, 2015–2019.

Research Supervisors: Noah Smith & Mari Ostendorf

Selected Publications

- [1] **M. Heddaya**, R. Wadhawan, M. Roberts, C. Tan. [When Internalization Fails: Finding Better Targets for Reasoning Compression](#). Under review, *ACL 2026*.
- [2] **M. Heddaya**, K. MacMillan, A. Malani, H. Mei, C. Tan. [CaseSumm: A Large-Scale Dataset for Long-Context Summarization](#). *Findings of NAACL 2025*. Accepted with talk @ *ALEA 2024*.
- [3] **M. Heddaya**, C. Tan, R. Voigt, Q. Zeng, A. Zentefis. [A Century of Inflation Narratives](#). *SSRN 2025*.
- [4] **M. Heddaya**, Q. Zeng, C. Tan, R. Voigt, A. Zentefis. [Causal Micro-Narratives](#). *WNU @ EMNLP 2024*.
- [5] **M. Heddaya**, S. Dworkin, C. Tan, R. Voigt, A. Zentefis. [Language of Bargaining](#). *ACL 2023*.

Research

PhD Student, University of Chicago CS

2021–2026

- [ongoing] Built and deployed [CivicChats.org](#), an end-to-end AI platform for civic deliberation. Built CivicEval, an automated evaluation system whose outputs inform iterative prompt optimization over collected user data. Running a controlled experiment on voter understanding. Follow-up work on large-scale data curation, synthetic data generation, and training to elicit coherent attitudes in models beyond surface style. [\[blog\]](#)
- Demonstrated that chain-of-thought internalization does not scale to complex reasoning [1]; proposed post-think distillation, a new compression target that preserves the teacher’s solution path and achieves the best accuracy–efficiency trade-off on competition math.
- Developed a method for at-scale automatic extraction of causal narratives and applied it to 4.2M sentences from a century of U.S. inflation coverage [3]; found causal narratives predict consumer expectations and spread via contagion and differentiation.
- Built a 25.6K Supreme Court opinion–summary benchmark [2]; showed fine-tuning improves automatic metrics while increasing hallucination rate. Frontier models hallucinate less but more subtly, requiring expert review to detect. Neither standard metrics nor LLM-as-judge reliably catch these failures.
- Collected ~400 human **bargaining** interactions with real incentives [5]; language reduces price variance $3\times$ and simple linguistic features outperform transformers at predicting outcomes.

ML Research Scientist Intern, [Abridge AI](#)

Summer 2025

- Developed methods for **reasoning compression** [1] in language models, identifying when internalization fails and finding better targets for distilling reasoning.

Applied Scientist Intern, [Amazon AWS AI Labs \(Bedrock\)](#)

Summer 2023

- Developed self-supervised alignment, a method for aligning LLMs to human preferences without RLHF, removing the need for reward models and reducing annotation requirements.

- The model scores its own outputs and uses self-feedback during fine-tuning as a regularizer for alignment.

Research Engineer, University of Washington

2020–2021

- Industry collaboration building unsupervised & supervised information extraction systems for conversational speech data (Advisors: Noah Smith & Mari Ostendorf). [\[paper\]](#)

Other Papers & Research Mentorship

- J. Fu, M. Heddaya, C. Tan. [Automatically Generating Hard Math Problems from Hypothesis-Driven Error Analysis](#). *LLM Reasoning @ ICLR 2026*.
- C. Shah, A. Agarwal, K. Garg, M. Heddaya. [LLM Rationalis? Measuring Bargaining Capabilities of AI Negotiators](#). *MTI @ NeurIPS 2025*.

Invited Talks

[Freestone Grove Partners](#), April 2025 — Causal Micro-Narratives.

[Max Planck Institute for Research on Collective Goods](#), February 2025 — NLP in the Legal Domain.

[University of Chicago LEAP Workshop](#), January 2023 — Language of Bargaining.

Service

- [ARR](#) Reviewer: June 2024, November 2024, October 2025, January 2026.
- [IC2S2](#) Reviewer: 2022, 2023, 2024, 2025, 2026.
- Program Committee, [Symposium on Human + AI](#), Fall 2022.

Teaching

University of Chicago

CMSC 25400 – *Machine Learning*, Winter 2023

CMSC 25300 / 35300 – *Mathematical Foundation of Machine Learning*, Fall 2022

CMSC 35100 – *Natural Language Processing*, Winter 2022

Skills & Methods

- **Technical:** Python, PyTorch, HuggingFace transformers/trl, LoRA/PEFT, vLLM, Node.js, PostgreSQL, full-stack development.
- **Research:** LLM fine-tuning & prompting, RLHF/alignment, experimental design, survey design, evaluation frameworks, large-scale text analysis.